

Executive Summary: Prostatic enlargement (benign prostatic hyperplasia, BPH) is highly prevalent in aging men – affecting ~50% by age 60 and over 80% by age 80 ¹ ² – and its global burden is rising (≈ 112 million cases worldwide in 2021) ³. Known risk factors include non-modifiable ones (age, genetics) and modifiable factors (metabolic syndrome, obesity, sedentary lifestyle, diet, etc.) ⁴ ⁵. Early prediction of BPH development could enable timely intervention, but validated risk scores are scarce. Recent efforts have produced models such as a LightGBM-based nomogram incorporating age, hypertension duration, glucose, urate and creatinine ($AUC \approx 0.69$) ⁶, and a smaller nomogram using fPSA, IL-8, and nesfatin-1 ($AUC \approx 0.85$) ⁷. Standard management (lifestyle measures, α -blockers, 5 α -reductase inhibitors (5-ARIs), etc.) is well-established ⁸ ⁹. This study will develop and validate a BPH risk prediction model in adult men, using current evidence on predictors and contemporary management guidelines.

1.0 Introduction

Benign prostatic hyperplasia (BPH) – nonmalignant enlargement of the prostate – is the leading cause of lower urinary tract symptoms (LUTS) in older men ¹⁰. Prevalence rises markedly with age ¹ ¹¹: roughly 20–50% of men in their 50s–60s and >80% of men in their 80s are affected ¹ ¹¹. Recent estimates indicate 112 million cases worldwide in 2021 (vs. 50 million in 1990) ³. BPH causes irritative (frequency, urgency, nocturia) and obstructive (weak stream, retention) symptoms that degrade quality of life ¹² ¹³. It can lead to severe complications (acute retention, infections, renal insufficiency) and imposes a large economic burden (e.g. US\$1.1B/year in US in 2000) ¹³. Although current management guidelines emphasize symptom-based therapy ⁸ ¹⁴, there is growing recognition (e.g. by WHO and urological associations) of the need to identify high-risk individuals for preventive intervention ¹⁵.

1.1 Background of Study

Age is by far the strongest risk factor for BPH; family and twin studies suggest heritability accounts for ~40–70% of cases ¹⁶. Modifiable risk factors have also been identified. Obesity, insulin resistance and metabolic syndrome are consistently linked to BPH risk and progression ⁴ ⁵. For example, dyslipidemia, hypertension, and hormonal imbalance associated with metabolic syndrome may promote prostatic growth ⁴. Diet and lifestyle also play roles: high animal protein intake, sedentary lifestyle, and excessive alcohol or smoking have been implicated in BPH development ⁴ ⁵. In contrast, diets rich in fruits/vegetables and regular exercise appear protective ⁵. Several urinary and serum biomarkers (e.g. PSA density, inflammatory cytokines, urinary metabolites) are under investigation, but none are yet established for risk prediction.

Numerous predictive models for BPH onset have been proposed in recent years. For example, Zi et al. (2025) developed a machine-learning model using UK Biobank data ($n \approx 210,000$) and identified five key predictors (age, hypertension duration, blood glucose, serum urate, creatinine) to estimate incident BPH ($AUC \approx 0.69$) ⁶. Other studies have explored nomograms: one used only patients with metabolic syndrome ($n=185$) to predict BPH with a C-index of 0.857, but lacked external validation ¹⁷. A cohort study in China (CHARLS, $n=3310$) examined circadian rhythm and metabolic variables for BPH risk, but achieved only modest AUC (~ 0.57) ¹⁸. A recent preprint developed a biomarker-based nomogram (fPSA, IL-8,

Nesfatin-1) with strong performance (AUC~0.85) in a small Chinese cohort ⁷. Table 1 compares key models: those based on large population cohorts (e.g. UK Biobank) tend to use common clinical/lab predictors, while some smaller studies incorporate novel biomarkers. External validation is generally lacking except in the largest model ⁶.

Model (Author, Year)	Predictors	Population (Follow-up)	Performance (AUC/C- index)	Validation
LightGBM nomogram (Zi et al., 2025) ⁶	Age, hypertension duration, blood glucose, serum urate, creatinine (5 variables; derived from 17)	UK Biobank (n=210,408 men, median 13.2y)	AUC ≈0.688 ⁶	Internal 5-fold CV; external in China (CHARLS n=5394, Fengshen n=294)
Nomogram (Liang et al., 2025) ⁷	fPSA, IL-8, Nesfatin-1	Chinese men (n=150; 88 BPH vs 62 controls)	AUC_train=0.843; AUC_val=0.856 ⁷	Internal train/ validation only
Circadian/ MetS model (Lv et al., 2023) ¹⁸	Circadian Rhythm Syndrome, Metabolic Syndrome	CHARLS China (n=3310 men, 4y)	AUC_circadian=0.574; AUC_MetS=0.561 ¹⁸	None

flowchart TD

A[Study population: adult men] --> B[Baseline data collection
(demographics, labs, lifestyle)]

B --> C[Predictor selection
(clinical factors, biomarkers)]

C --> D[Model development
(training set)]

D --> E[Internal validation
(test set, CV)]

E --> F[Performance evaluation
(AUC, calibration)]

F --> G[External validation
(independent cohort)]

G --> H[Model refinement & web tool]

1.2 Statement of Problem

Despite its prevalence and impact, BPH often remains unrecognized until significant symptoms arise. Current evaluation focuses on symptom grading (IPSS) and examination rather than proactive risk assessment ¹⁹ ¹⁵. There is no widely adopted tool to predict incident BPH in asymptomatic men. Existing studies have been limited by small samples, short follow-up, and lack of diverse populations ²⁰. Furthermore, most models remain confined to research settings (e.g. requiring advanced biomarkers or Chinese cohorts) and have not been implemented in practice. This gap hinders early identification of men who might benefit from lifestyle modification or chemoprevention. Our study addresses this by developing a broadly applicable risk prediction model using readily available predictors and validating it externally, thereby enabling earlier intervention in primary care.

1.3 Aim and Objectives of the Study

Aim: To develop and validate a multivariable risk prediction model for incident prostate enlargement (BPH) in adult men.

Objectives: (1) Identify candidate predictors of BPH (demographics, comorbidities, lifestyle, laboratory/biomarker data) from literature and cohort data. (2) Construct prediction algorithms (statistical and machine-learning) using a large training cohort. (3) Assess model performance (discrimination, calibration) internally and tune via cross-validation. (4) Externally validate the model in independent cohorts (geographically/ethnically distinct). (5) Compare new model performance to existing ones and produce a user-friendly risk calculator.

1.4 Significance of the Study

This study is significant because it addresses a high-burden condition by enabling pre-emptive risk stratification. BPH imposes large costs (e.g. ~\$1.1B USD direct costs in 2000) and reduces quality of life ²¹. Identifying high-risk men early could allow tailored counseling (e.g. diet, exercise) or medical prevention (α -blockers or 5-ARIs in early disease) to delay progression ⁸ ⁹. A reliable risk model would aid physicians and patients in shared decision-making, focusing resources on those most likely to benefit. It aligns with global NCD prevention efforts by integrating urologic disease into chronic disease management frameworks ¹⁵. Finally, understanding the predictors of BPH advances scientific knowledge and may suggest novel prevention strategies.

1.5 Scope of the Study

This study focuses on **adult men** (age ≥ 40) without prior prostate disease. The model will predict new-onset BPH (prostate enlargement) over a multi-year horizon. The geographic scope is unspecified (global intent); we will use diverse cohort data but acknowledge variation by ethnicity. Predictors will be limited to routinely available clinical and laboratory measures. Outcomes are defined by clinical diagnosis of BPH (ICD codes or ultrasound volume ≥ 30 mL). Surgical techniques, neurological causes of LUTS, or prostate cancer are beyond the scope. The study emphasizes risk prediction; comprehensive evaluation of treatment outcomes (beyond risk factors) is not included here.

1.6 Limitations of the Study

Potential limitations include cohort selection bias (e.g. healthier volunteer bias in biobank data), which may reduce generalizability. Some predictors (e.g. self-reported sleep disturbance) may be imprecise. The model may overfit if too many predictors are used; we will mitigate this via penalization and validation. Ethnic and regional differences in BPH risk may limit applicability outside studied populations ²². Additionally, the model's performance depends on data quality and consistency (e.g. diagnostic criteria across cohorts). Ethical considerations are paramount: we will use de-identified data under IRB approval, maintain patient privacy, and address algorithmic bias by including diverse subgroups. The model should be used to

complement (not replace) clinical judgment to avoid overdiagnosis. We will follow reporting guidelines (TRIPOD) and ensure transparency of methods.

Table 1: Comparison of published BPH risk prediction models 6 7 18 .

Figure 1: Proposed study design (flowchart) for developing and validating the BPH risk prediction model.

Table 1: See above for comparison of key predictive models (Zi et al. 2025; Liang et al. 2025; Lv et al. 2023).

References:

1. Arnold MJ, Gaillardetz A, Ohiokpeha J. Benign prostatic hyperplasia: Rapid Evidence Review. *Am Fam Physician*. 2023;107(6):613–622.
2. Chen X, Yang S, He Z, Tang X, Gong Y, Zhang X, et al. Comprehensive analysis of global, regional, and national burden of benign prostatic hyperplasia from 1990 to 2021. *Sci Rep*. 2025;15:5644.
3. Siqueira MHB. Risk factors for benign prostatic hyperplasia: a comprehensive review. *Rev Assoc Med Bras (1992)*. 2025;71(6):e20250343.
4. Yoosuf BT, Panda AK, Favas KT, Bharti SK, Devana SK, et al. Comparative efficacy and safety of α -blockers as monotherapy for benign prostatic hyperplasia: a systematic review and network meta-analysis. *Sci Rep*. 2024;14:11116.
5. Zi H, Wang YB, Huang Q, Zhang YY, He FZ, Xing LM, et al. Predicting benign prostatic hyperplasia risks: model development and external validation based on three cohorts. 2025. (preprint)
6. Liang YL, Wu JH, Wu Q, Luo HD, Huang KK, Yan YH, Yi BL. Establishment of a risk prediction model for benign prostatic hyperplasia donors: a retrospective cohort study in China. *SSRN Preprint*. 2025.

1 4 Comparative efficacy and safety of alpha-blockers as monotherapy for benign prostatic hyperplasia: a systematic review and network meta-analysis - PMC
<https://pmc.ncbi.nlm.nih.gov/articles/PMC11096304/>

2 3 Comprehensive analysis of the global, regional, and national burden of benign prostatic hyperplasia from 1990 to 2021 | Scientific Reports
https://www.nature.com/articles/s41598-025-90229-3?error=cookies_not_supported&code=613e8a39-cc69-437d-8964-f5287c40b559

5 11 16 Risk factors for benign prostatic hyperplasia: a comprehensive review - PMC
<https://pmc.ncbi.nlm.nih.gov/articles/PMC12245072/>

6 10 13 15 17 18 20 21 22 Predicting benign prostatic hyperplasia risks: model development and external validation based on three cohorts - PMC
<https://pmc.ncbi.nlm.nih.gov/articles/PMC12709829/>

7 Establishment of a risk prediction model for benign prostatic hyperplasia donors: a retrospective cohort study in China by Yunlai Liang, Jiahui Wu, Qiong Wu, Huidan Luo, Kangkang Huang, Yahui Yan, Bin Yi :: SSRN
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5957096

8 19 Benign Prostatic Hyperplasia: Rapid Evidence Review | AAFP
<https://www.aafp.org/pubs/afp/issues/2023/0600/benign-prostatic-hyperplasia.html>

9 14 Pharmacological Approaches in Managing Symptomatic Relief of Benign Prostatic Hyperplasia: A Comprehensive Review - PMC

<https://pmc.ncbi.nlm.nih.gov/articles/PMC10823302/>

12 Untargeted metabolomics revealed urinary metabolic pattern for discriminating prostate cancer from benign prostatic hyperplasia in Chinese participants - PMC

<https://pmc.ncbi.nlm.nih.gov/articles/PMC12238017/>